



COMPETENCY STANDARD
FOR
ADVANCED MASONRY
(Construction Sector)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

COPYRIGHT

The Competency Standards for **Advanced Masonry** is designed to guide the development of curricula, teaching materials, and assessment tools. It also serves as a framework for providing industry-aligned training, ensuring that individuals meeting the prescribed standards through assessments are qualified to secure relevant employment opportunities.

This document has been specifically developed for training purposes under the Skills for Industry Competitiveness and Innovation Program (SICIP), aiming to address the skills requirements of the construction sector. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

While public and private institutions are permitted to use the information from this standard for activities that benefit Bangladesh, other parties must seek prior permission from the document's owner for reproducing its content, either partially or entirely, in English or other languages.

This document is available at:

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance,
Probashi Kallyan Bhaban (Level 15 & 16)
71-72 Old Elephant Road, Eskaton Garden
Dhaka-1000
Phone: +8802 55138753-55, Fax: +88 02 55138752
Website: www.sicip.gov.bd

INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Occupation Specific Competencies (120 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-CON-AM-01-O	Make Masonry Mortar/Stucco	1. Collect relevant tools, equipment and materials 2. Prepare mortar mixing box 3. Make mortar/stucco 4. Clean and maintain the work area	10
SICIP-CON-AM-02-O	Carryout Pavement Laying Work	1. Gather pavement laying tools, equipment and materials 2. Prepare cement concrete 3. Set up base for paving work 4. Lay paving bricks/blocks 5. Complete the paving work 6. Clean/maintain the workplace	10
SICIP-CON-AM-03-O	Perform Bricks and Block Works	1. Prepare work area 2. Build a concrete footer 3. Lay brick or block 4. Finish brick or block work 5. Perform curing 6. Clean/maintain the workplace	30
SICIP-CON-AM-04-O	Carry-Out Stone Works	1. Collect masonry working tools, equipment and materials 2. Organize stone at work site 3. Create cement mortar/stucco 4. Carryout stone installation work 5. Carryout wall paneling works with stone 6. Perform curing 7. Clean/maintain the workplace	20
SICIP-CON-AM-05-O	Perform Masonry Surface Plastering	1. Clean the masonry surface area prior to plastering 2. Mix plaster materials 3. Apply plaster on plain surfaces 4. Apply plaster to corners and roof 5. Clean/maintain the workplace	20
SICIP-CON-AM-06-O	Apply Waterproofing Activities in Construction	1. Organize work area for waterproofing 2. Mix waterproofing materials	15

		with concrete/plaster 3. Perform waterproofing works 4. Clean/maintain the workplace	
SICIP-CON-AM-07-O	Understand Modern Masonry Techniques	1. Understand ready mix casting techniques 2. Describe auto plastering techniques 3. Explain auto brick laying techniques with robot	15
Total Hour			120

COMPETENCY STANDARD: ADVANCED MASONRY

A. Occupation Specific Competencies

Unit of Competency: MAKE MASONRY MORTAR/STUCCO	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-AM-1-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required for a worker to make masonry mortar/stucco. It specifically includes the tasks of collecting relevant tools, equipment and materials, preparing mortar mixing box, making mortar/stucco and cleaning and maintaining the work area.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collect relevant tools, equipment and materials	1.1 <u>Personal Protective Equipment (PPE)</u> are selected and used. 1.2 <u>Tools, equipment & materials</u> are selected and gathered properly. 1.3 Tools, equipment & materials are checked for usability and quality.
2. Prepare mortar mixing box	2.1 Mortar mixing box is cleaned. 2.2 Mixing box is positioned on a stable and level surface. 2.3 Mortar box is checked.
3. Make mortar/stucco	3.1 Sand and cement ratio are maintained and measured. 3.2 Materials are laid on mortar mixing box as per instruction. 3.3 Sand, cement and water are mixed in accordance with specification. 3.4 <u>Transport</u> is used to carry materials at the working place.
4. Clean and maintain the work area	4.1 Tools & equipment are cleaned and stored in its proper storage. 4.2 Mixing box/containment is cleaned. 4.3 Waste materials are disposed in proper place.

Range of Variables

Variable	Range
	May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Safety helmet 1.2 Safety shoes 1.3 Hand gloves 1.4 Safety apron 1.5 Safety mask

2. Tools, equipment & materials	2.1 Tools; 2.1.1 Measuring box 2.1.2 Sand sieve 2.1.3 Shovel 2.1.4 Masonry tray 2.1.5 Bucket 2.1.6 Mug 2.2 Equipment 2.2.1 Mixer 2.2.2 Mixing box 2.3 Materials 2.3.1 Cement 2.3.2 Sand 2.3.3 Water
3. Transport	3.1 Masonry tray: plastic/metal 3.2 Trolley 3.3 Wheel borrows

Curricular Content Guide

1. Underpinning Knowledge	1.1 Tools, equipment, and materials for mortar mixing tasks. 1.2 Measuring procedures of sand and cement ratios. 1.3 Mixing process of sand and cement.
2 Underpinning Skills	2.1 Selecting and using PPE 2.2 Selecting and gathering tools, equipment & materials 2.3 Preparing platform area as per workplace requirements 2.4 Maintaining sand and cement ratio measurement 2.5 Mixing sand, cement and water are maintained 2.6 Cleaning tools & equipment 2.7 Cleaning platform area 2.8 Disposing waste materials in designated place
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4 Resource Implications	4.1 Workplace (simulated or actual) 4.2 Mortar preparation equipment, tools and materials 4.3 Work instruction sheet

Assessment Evidence Guide

Advanced Masonry

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collected relevant tools, equipment and materials. 1.2 prepared mortar mixing box. 1.3 make mortar/stucco. 1.4 cleaned and maintained the work area.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: CARRYOUT PAVEMENT LAYING WORK	Nominal Duration: 10 hrs.	Unit Code: SICIP-CON-AM-2-O
--	-------------------------------------	---------------------------------------

Unit Descriptor:

This unit covers the knowledge, skills and attitudes required for a worker to carry-out pavement laying work in construction sector. It specifically includes the tasks of gathering pavement laying tools, equipment and materials, preparing raw materials for concrete making, setting up base for paving work, laying paving bricks/blocks and completing the paving work.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather pavement laying tools, equipment and materials	1.1 <u>Personal Protective Equipment (PPE)</u> is selected and used. 1.2 <u>Tools and equipment</u> are gathered, checked and prepared. 1.3 Bricks, paving tiles/blocks are selected, collected and gathered at work site. 1.4 Mortar/grouting materials are selected, collected and gathered.
2. Prepare cement concrete	2.1 Site for mixing of <u>raw materials</u> is identified and located. 2.2 Tools and equipment are carried out to the work site. 2.3 Brick/paving tiles or blocks are cut into specified sizes by power saw or paving tile cutters. 2.4 Mortar/grout making raw materials are carried to work site.
3. Set up base for paving work	3.1 Base is selected in accordance with building plan. 3.2 Unnecessary materials are removed from the base site. 3.3 Forms for pavement making are built in accordance with workplace requirement. 3.4 Appropriate type of sand is placed on the base in accordance with workplace requirements. 3.5 Base is leveled and tamped in accordance with workplace specification.
4. Lay paving bricks/blocks	4.1 Paving line is set and aligned in the work site. 4.2 Paving line and perpendicular lines are set up at two ends of the line. 4.3 Paving bricks/blocks are positioned and leveled.
5. Complete the paving work	5.1 Gap between blocks or bricks are filled up with appropriate <u>fillers</u> . 5.2 Final level check is made and correction is carried out where necessary. 5.3 Unnecessary materials are removed and cleaned from blocks or bricks. 5.4 Curing time is observed before use of the newly built pavement.
6. Clean/maintain the	6.1 Tools and equipment are cleaned.

workplace	6.2 Workplace is cleaned. 6.3 Waste materials are disposed in its designated/proper place.
-----------	---

Range of Variables

Variable	Range May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Skull guard/helmet 1.2 Dust mask 1.3 Goggles 1.4 Safety shoes 1.5 Apron 1.6 Hand Gloves
2. Tools and equipment	2.1 Tools 2.1.1 Measuring Tape (30m) 2.1.2 Tri-square 2.1.3 Pocket tape (3m) 2.1.4 Claw hammer/crow bar 2.1.5 Shovel 2.1.6 Center pins 2.1.7 Water tube level 2.1.8 Water level (precision) 2.1.9 Masonry Trowel 2.1.10 Concrete pans 2.2 Equipment 2.2.1 Trolley 2.2.2 Van 2.2.3 Mixing board 2.2.4 Concrete mixing machine 2.2.5 Tiles, brick and block cutter machine
3. Raw materials	3.1 Cement 3.2 Sand 3.3 Gravel 3.4 Water 3.5 Wood forms 3.6 Metal forms 3.7 Plastic forms 3.8 Cotton rags
4. Fillers	4.1 Sand 4.2 Cement grout 4.3 Polyurethane concrete crack sealant 4.4 Polymer based sealers

Curricular Content Guide

1. Underpinning Knowledge	1.1 Different types of paving work 1.2 Quality of materials used for paving work 1.3 Tools, equipment and machinery used for paving work 1.4 Procedure of setting blocks/tiles for paving work 1.5 Methods and techniques of performing paving works 1.6 Different types of filler materials for grouting/filling of gaps between blocks/tiles. 1.7 Procedure of preparing and filling gaps
2. Underpinning Skills	2.1 Selecting, collecting and gathering bricks, paving tiles and blocks at work site 2.2 Gathering tools, equipment and materials to prepare for paving 2.3 Cutting brick, paving tiles and blocks into specified sizes by power saw or paving tile cutters 2.4 Mixing cement, sand and water for grouting using appropriate ratio 2.5 Levelling base according to specifications 2.6 Setting-up paving line and aligning in the work site 2.7 Placing and levelling paving blocks/tiles 2.8 Preparing and using cement grouting as per instruction 2.9 Filling-up gaps between blocks or bricks/tiles with appropriate fillers 2.10 Cleaning tools, equipment and worksite 2.11 Disposing waste materials in its designated/proper place
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of paving work tools and equipment 4.3 Paving work materials and consumables 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Paving work instruction manual

Assessment Evidence Guide

1. Critical Aspects of	Assessment required evidence that the candidate ensures:
------------------------	--

Competency	1.1 gathered pavement laying tools, equipment and materials. 1.2 prepared cement concrete. 1.3 set up base for paving work. 1.4 laid paving bricks/blocks. 1.5 completed the paving work. 1.6 cleaned/maintained the workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM BRICKS AND BLOCK WORK	Nominal Duration: 30 hrs.	Unit Code: SICIP-CON-AM-3-O
Unit Descriptor:		

This unit covers the knowledge, skills and attitudes required for a worker to pile structural bricks and blocks to build structures for construction. It specifically includes the tasks of preparing the location, building a concrete footer, laying brick or block structure, finishing brick or block laying work and curing and cleaning/maintaining the workplace.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare work area	1.1 Plans are read and interpreted. 1.2 <u>Personal Protective Equipment (PPE)</u> is collected and used. 1.3 <u>Tools and equipment</u> are selected and prepared. 1.4 <u>Materials</u> are selected and prepared. 1.5 Building lines are located in accordance with building plan. 1.6 <u>Types of bricks and blocks</u> are identified. 1.7 Horizontal and vertical guide for brick and block is installed.
2. Lay brick or block	2.1 Bricks and blocks are laid on the line at minimum allowance of 1/16 inch. 2.2 Mortars are spread and filled on the base and gaps of bricks and blocks. 2.3 Bricks and blocks are positioned and laid. 2.4 Bricks and blocks are leveled using appropriate levelling device. 2.5 Pointing of brick joints is performed.
3. Finish brick or block work	3.1 Excess mortar is removed from the surface of the bricks or blocks using a trowel. 3.2 Joints between bricks or blocks are finished by compacting the mortar. 3.3 Surface alignment is checked for consistency. 3.4 Top surface is cleaned and finished. 3.5 Completed work is reported for final checking.
4. Perform curing	4.1 <u>Curing methods</u> are selected based on the type of materials and environmental conditions. 4.2 Curing is applied, ensuring the surface of the mortar. 4.3 Curing duration is monitored to ensure that the material is kept moist for the required period, typically 7-14 days. 4.4 Temperature and humidity levels are monitored. 4.5 Curing is periodically checked.
5. Clean/maintain the workplace	5.1 Tools and equipment are cleaned. 5.2 Work place is cleaned in accordance with workplace requirements. 5.3 Waste materials are disposed in proper place.

Range of Variables

Variable	Range May include but not limited to:
1. Personal Protective Equipment (PPE)	1.1 Skull guard/helmet 1.2 Dust mask 1.3 Goggles 1.4 Safety shoes 1.5 Safety hand gloves 1.6 Safety apron 1.7 Body harness/safety belt
2. Tools & equipment	2.1 Tools 2.1.1 Measuring Tape (30m) 2.1.2 Tri-square 2.1.3 Pocket tape (3m) 2.1.4 Claw hammer/crow bar 2.1.5 Shovel 2.1.6 Center pins 2.1.7 Water tube level 2.1.8 Water level (precision) 2.1.9 Plumb bob 2.1.10 Masonry Trowel 2.1.11 Brick hammer 2.1.12 Chipping hammer 2.1.13 Pointing trowel 2.1.14 Masonry tray 2.2 Equipment 2.2.1 Brick cutting machine 2.2.2 Concrete mixing board
3. Materials	3.1 Cement 3.2 Sand 3.3 Gravel 3.4 Paving brick 3.5 Blocks
4. Types of bricks and blocks	4.1 Bricks 4.1.1 Handmade bricks 4.1.2 Auto/ceramic bricks (hollow and solid bricks) 4.1.3 Cements sand bricks 4.2 Blocks 4.2.1 Hollow blocks 4.2.2 Solid blocks (aerated autoclaved, cellular lightweight, expanded)
5. Building Plan	5.1 Civil Drawings 5.2 Scale 5.3 Measurement

	5.4 Electrical plans
	5.5 Architectural plans
	5.6 Masonry plan
6. Curing methods	6.1 Water spraying
	6.2 Fog misting
	6.3 Saturated coverings

Curricular Content Guide

1. Underpinning Knowledge	1.1 Appropriate tools, equipment, and materials needed for brick and block laying. 1.2 Locate and mark building lines in accordance with building plans. 1.3 Identify various types of bricks and blocks. 1.4 Horizontal and vertical guides. 1.5 Lay bricks and blocks on marked lines, maintaining a minimum allowance of 1/16 inch for precision. 1.6 Spread and fill mortar on the base and in gaps between bricks and blocks to ensure proper bonding. 1.7 Position and lay bricks and blocks carefully to achieve structural integrity. 1.8 Level bricks and blocks using appropriate leveling devices to ensure even surfaces. 1.9 Checking procedure of surface alignment. 1.10 Curing processes to keep mortar surfaces moist and facilitate proper setting.
2. Underpinning Skills	2.1 Locating building lines 2.2 Establishing location of brick structure and block structure based on reference lines of building for proper alignment and dimension 2.3 Installing horizontal and vertical guide for brick and block 2.4 Marking the lay-out of brick and block structure correctly 2.5 Laying bricks and blocks on the line at minimum allowance of 1/16 inch and installing reinforcing bar 2.6 Spreading and filling mortars on the base and gaps of bricks and blocks 2.7 Checking plumb and level constantly during brick and block laying 2.8 Making final checks to ensure that work is accurate 2.9 Ensuring that curing is done properly 2.10 Cleaning work area and storing, tools and equipment in its designated storage area
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment 4.3 Materials and consumables 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Instruction manual

Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 prepared work area. 1.2 laid brick or block. 1.3 finished brick or block work. 1.4 performed curing. 1.5 cleaned/maintained the workplace.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: CARRY-OUT STONE WORKS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-CON-AM-4-O
Unit Descriptor:		

This unit covers the knowledge, skills and attitudes required for a worker to carry-out stone works and brick works. It specifically includes the tasks of gathering masonry working tools, equipment and materials at site, organizing bricks/blocks at work site, creating cement mortar/stucco, preparing for brick/block installation, carrying-out brick/ block installation works and cleaning/maintaining the workplace.

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Gather masonry tools, equipment and materials	1.1 <u>PPE</u> is selected & used. 1.2 Masonry working <u>tools</u> and <u>materials</u> are gathered and checked for usability/quality. 1.3 Transport is used for carrying materials. 1.4 Scaffolding is set for masonry works.
2. Organize stone at work site	2.1 Stones are cleaned and soaked in accordance to specification. 2.2 Quality of stones are checked and segregated. 2.3 Stones are organized at work site in accordance to workplace requirements.
3. Create cement mortar/stucco	3.1 Quality of cement, sand, stone, and water is determined by assessing their condition. 3.2 Ratio of sand, stone, cement, and water is determined in accordance with the <u>construction plan</u> and specifications. 3.3 Materials (sand, stone, cement, and water) are mixed according to the workplace requirements and specifications. 3.4 Mortar or stucco is tested for consistency by checking its workability. 3.5 Mixed mortar or stucco is inspected for any inconsistencies in texture.
4. Carryout stone installation work	4.1 Area for stone installation is prepared. 4.2 Stone materials are selected according to the design specifications. 4.3 Mortar bed is prepared and applied to the installation surface. 4.4 First row of stones is laid carefully using a level to ensure accuracy. 4.5 Stone placement is carried out ensuring that the joints are uniform and the overall structure is stable. 4.6 Stone cutting is done as needed to fit stones into corners, edges, or other irregular shapes, ensuring clean. 4.7 Stone joints are filled with mortar, ensuring they are fully packed.

5. Carry out wall paneling works with stone	5.1 Wall surface is prepared by ensuring it is clean, level, and free from debris. 5.2 Stone panels are selected according to the design specifications. 5.3 The stone is soaked for a minimum time of 12-24 hours. 5.4 Layout for the stone panels is marked on the wall. 5.5 Mortar is mixed and applied to the back of the stone panels. 5.6 Edges of the panels are checked for alignment. 5.7 Joints between the stone panels are filled with mortar or grout. 5.8 Excess mortar is removed from the surface of the stone panels.
6. Perform curing	6.1 Curing methods are selected based on the type of materials and environmental conditions. 6.2 Curing is applied, ensuring the surface of the mortar. 6.3 Curing duration is monitored to ensure that the material is kept moist for the required period, typically 14-28 days. 6.4 Temperature and humidity levels are monitored. 6.5 Curing is periodically checked.
7. Clean/maintain the workplace	7.1 Tools and equipment are cleaned. 7.2 Work place is cleaned. 7.3 Waste materials are disposed in proper place.

Range of Variables

Variable	Range May include but not limited to:
1. PPE	1.1 Safety helmet 1.2 Dust mask 1.3 Safety goggles 1.4 Safety shoes 1.5 Safety gloves 1.6 Safety apron 1.7 Body harness/Safety belt
2. Tools	2.1 Tools 2.1.1 Measuring Tape (30m) 2.1.2 Tri- square 2.1.3 Pocket tape (3m) 2.1.4 Claw hammer/crow bar 2.1.5 Shovel 2.1.6 Center pins 2.1.7 Water tube level 2.1.8 Water level (precision) 2.1.9 Masonry Trowel

	2.1.10 Point trowel 2.1.11 Concrete pans
3. Materials	3.1 Cement 3.2 Sand 3.3 Gravel 3.4 Stone chips 3.5 Clean Water
4. Construction plan	4.1 Civil Drawings 4.2 Scale 4.3 Measurement specification 4.4 Electrical plans 4.5 Architectural plans 4.6 Masonry plan/ Layout Plan

Curricular Content Guide

1. Underpinning Knowledge	1.1 Set up procedure of scaffolding. 1.2 Quality and condition of cement, sand, stone. 1.3 Calculating methods of ratio sand, stone, cement, and water. 1.4 Mix procedures of mortar or stucco. 1.5 Cutting process of stones. 1.6 Filling procedures of stone joints with mortar. 1.7 Marking layout lines for stone panel placement. 1.8 Checking process of edges and joints. 1.9 Curing procedure of moisture on mortar surfaces.
2. Underpinning Skills	2.1 Gathering and checking masonry tools and materials for usability and quality 2.2 Setting scaffolding for masonry works 2.3 Mixing materials (sand, stone, cement, and water) according to workplace requirements and specifications 2.4 Preparing area for stone installation 2.5 Preparing and applying mortar bed to the installation surface 2.6 Cutting stones to fit corners, edges, or irregular shapes with clean finishes 2.7 Marking layout for stone panels on the wall 2.8 Filling joints between stone panels with mortar or grout 2.9 Cleaning tools and equipment 2.10 Disposing waste materials properly
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns

	3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of tools and equipment 4.3 Materials and consumables 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 gathered masonry tools, equipment and materials. 1.2 organized stone at work site 1.3 created cement mortar/stucco 1.4 carried out stone installation work 1.5 carried out wall paneling works with stone 1.6 performed curing 1.7 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: PERFORM MASONRY SURFACE PLASTERING	Nominal Duration: 20 Hrs.	Unit Code: SICIP-CON-AM-5-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required for a worker to accomplish masonry surface plastering for construction. It specifically includes the tasks of cleaning the masonry surface area prior to plastering, mixing plaster materials, applying plaster on plain surfaces, applying plaster to corners and cleaning/maintaining the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Clean the masonry surface area prior to plastering	1.1 Appropriate <u>PPE</u> is gathered and used. 1.2 Scaffolding is prepared as required. 1.3 <u>Tools, equipment</u> and <u>materials</u> are selected and prepared. 1.4 Masonry surface is inspected for dirt, dust, grease, or any other contaminants. 1.5 Surface is cleaned and washed for plastering. 1.6 Cracks or holes are filled and repaired. 1.7 Excess mortar or paint is scraped off from the masonry surface.
2. Mix mortar for plastering	2.1 <u>Foreign materials</u> and larger particles are separated from sand with the help of sieve/screen. 2.2 The required materials for mortar are gathered and measured according to the specified mix ratio for plastering. 2.3 Cement and sand are mixed in the correct proportion. 2.4 Water is added gradually to the dry mix while stirring continuously to achieve the desired consistency. 2.5 Mortar mix is tested for consistency by checking its workability. 2.6 Consistency of the mortar is checked.
3. Apply plaster on plain surfaces	3.1 Surface is lightly moistened with water. 3.2 Plastering tools are prepared and checked for proper functionality. 3.3 Plaster is applied in an even coat, starting from the top of the wall or surface. 3.4 Plaster layer is spread with consistent pressure, filling in gaps, seams, and joints while plaster is still soft. 3.5 Wooden trowel and wetted foam are used to finish the surface.

	3.6 Excess plaster is removed using broom and foam. 3.7 Plaster levelled is checked using levelling tools. 3.8 Plastered surface is checked for smoothness, evenness, and proper adhesion.
4. Apply plaster to corners and roof	4.1 Corners and roof surfaces are inspected. 4.2 <u>Corner plastering tools</u> are collected and checked for proper functionality. 4.3 Scaffolding is prepared as required. 4.4 Corners are plastered by applying a thin, even layer of plaster using a corner trowel. 4.5 Roof area is plastered by applying an even coat of plaster with a large trowel. 4.6 Plaster is smoothed and leveled using a float or trowel. 4.7 Finished plastered is cured in accordance to workplace specification. 4.8 Final inspection is performed on the corners and roof.
5. Clean/maintain the workplace	5.1 Tools and equipment are cleaned. 5.2 Work place is cleaned. 5.3 Waste materials are disposed in proper place.

Range of Variables

Variable	Range May include but not limited to:
1. PPE	1.1 Skull guard/helmet 1.2 Dust mask 1.3 Goggles 1.4 Safety shoes 1.5 Safety gloves 1.6 Proper working clothes 1.7 Body harness/Safety belt
2. Tools & equipment	2.1 Tools 2.1.1 Measuring Tape (30m) 2.1.2 Tri- square 2.1.3 Pocket tape (3m) 2.1.4 Claw hammer / crow bar 2.1.5 Shovel 2.1.6 Water tube level 2.1.7 Water level (precision) 2.1.8 Masonry Trowel 2.1.9 Point trowel 2.1.10 Concrete pans 2.2 Equipment 2.2.1 Truck

	2.2.2 Trolley 2.2.3 Pick up 2.2.4 Van 2.2.5 Mixing platform
3. Materials	3.1 Cement 3.2 Sand 3.3 Water 3.4 Cotton rags 3.5 Concrete
4. Foreign materials	4.1 Plastic 4.2 Paper 4.3 Leaves 4.4 Stone 4.5 Wood Scrap
5. Corner plastering tools	5.1 Plaster hawk 5.2 Wooden flat trowel 5.3 Metal flat trowel 5.4 Corner trowel 5.5 Pointed trowel

Curricular Content Guide

1. Underpinning Knowledge	1.1 Safe work practices and first aid regulations 1.2 Materials staging procedure 1.3 Chipping and racking methods/techniques 1.4 Surface sweeping and watering 1.5 Mortar preparation 1.6 Plastering procedure and techniques 1.7 Plastered surface leveling procedure and techniques
2. Underpinning Skills	2.1 Preparing scaffolding as required 2.2 Selecting and preparing tools, equipment and materials 2.3 Staging materials properly and must be free from any foreign matters 2.4 Performing racking out of joints and chipping as required. 2.5 Preparing surface for plastering 2.6 Screening sand and cement mixture before applying water 2.7 Preparing mortar as per specified ratio of sand, cement and clean water 2.8 Applying distribution of mortar on the surface evenly and leveled 2.9 Curing plastering surface as required 2.10 Setting jamb 2.11 Preparing sill and set 2.12 Checking and adjusting right angle, acute angle and

	obtuse angles
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of tools and equipment 4.3 Materials and consumables 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Instruction manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 cleaned the masonry surface area prior to plastering 1.2 mixed mortar for plastering 1.3 applied plaster on plain surfaces 1.4 applied plaster to corners and roof 1.5 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: APPLY WATERPROOFING ACTIVITIES IN CONSTRUCTION	Nominal Duration: 15 hrs.	Unit Code: SICIP-CON-AM-6-O
Unit descriptor: This unit covers the knowledge, skills and attitudes required for a worker to apply waterproofing activities in construction. It specifically includes the tasks of organizing work area for waterproofing, preparing concrete prior to waterproofing, applying waterproofing material, performing other waterproofing considerations and cleaning/maintaining the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Organize work area for waterproofing	1.1 <u>PPE</u> is selected and used. 1.2 <u>Tools and equipment</u> required for the waterproofing process. 1.3 <u>Materials</u> appropriate to the work application are identified, obtained & prepared. 1.4 Environmental requirements are identified and conformed. 1.5 Work area is organized to ensure easy access to tools, equipment, and materials. 1.6 Inspection of the work area is done to ensure that all tools, materials, and safety requirements are in place.
2. Mix waterproofing materials with concrete/plaster	2.1 Waterproofing materials are gathered. 2.2 Concrete or plaster mix is prepared according to the required proportions. 2.3 Mixing tools are used to blend the waterproofing materials. 2.4 Mix is tested for consistency by checking its workability. 2.5 Mix is adjusted, if necessary, by adding more water or waterproofing material. 2.6 The mixed material is inspected for uniformity and proper bonding properties.
3. Perform waterproofing works	3.3 Surface to be waterproofed is inspected, cleaned, and prepared by removing any dirt, debris, or loose materials. 3.4 Area to be waterproofed is marked or outlined. 3.5 Waterproofing material is applied to the surface using the appropriate method. 3.6 Multiple coats of waterproofing material are applied if necessary. 3.7 Sealing materials are used around edges, fixtures, and penetrations. 3.8 Drying time is monitored to ensure that the waterproofing material sets properly. 3.9 Final inspection is carried out to ensure to the required

	standards.
4. Clean/maintain the workplace	4.1 Tools and equipment are cleaned and stored. 4.2 Work area is cleaned. 4.3 Waste materials are disposed in proper place

Range of Variables

Variable	Range May include but not limited to:
1. PPE	1.1 Skull guard/helmet 1.2 Dust mask. 1.3 Goggles. 1.4 Safety shoes. 1.5 Safety gloves 1.6 Proper working clothes 1.7 Body harness/Safety belt
2. Tools & equipment	2.1 Tools 2.1.1 Measuring Tape (30m) 2.1.2 Tri- square 2.1.3 Pocket tape (3m) 2.1.4 Hammer 2.1.5 Shovel 2.1.6 Cold chisel 2.1.7 Levelling hose 2.1.8 Spirit level 2.1.9 Masonry Trowel 2.1.10 Pointed trowel 2.1.11 Foam / Sponge 2.1.12 Concrete pans 2.1.13 Wooden float 2.1.14 Steel float 2.1.15 Brush 2.1.16 Nylon string 2.1.17 Plumb bob 2.2 Equipment 2.2.1 Sand blasting equipment 2.2.2 Mixing board 2.2.3 Edger / Grinder 2.2.4 Scaffolding / ladder 2.2.5 Truck 2.2.6 Concrete mixer 2.2.7 Concrete pump 2.2.8 Trolley 2.2.9 Pick up
3. Materials	3.1 Cement

	3.2 Concrete 3.3 Sand (screened) 3.4 Stone/brick chips 3.5 Waterproofing reagent 3.6 Water 3.7 Roof sealer
--	---

Curricular Content Guide

1. Underpinning Knowledge	1.1 Work instruction and operational detail 1.2 Signage and barricade identification 1.3 Drawings interpretation 1.4 Bond breaker application 1.5 Water proofing procedures and techniques 1.6 Potential faults identification techniques 1.7 Water proofing materials checking method and technique 1.8 Water proofing membrane preparation 1.9 Water proofing materials mixing
2. Underpinning Skills	2.1 Obtaining work instructions and operational details for relevant information, planning and preparation. 2.2 Selecting and gathering PPEs, waterproofing materials, tools and equipment 2.3 Identifying application of bond-breaker and waterproofing installation 2.4 Checking waterproofing material for conformity and compatibility with substrate material 2.5 Preparing substrates to be waterproofed 2.6 Prime coating substrates with waterproofing material. 2.7 Preparing flashings and ready for placement and fixing. 2.8 Preparing waterproofing membrane for installation. 2.9 Mixing or preparing waterproofing material 2.10 Applying waterproofing materials 2.11 Checking Quality of works and rectifying defects 2.12 Cleaning work area and storing tools and equipment
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of waterproofing work tools and equipment

	4.3 Waterproofing materials and consumables 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Waterproofing work instruction manual
--	--

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 organized work area for waterproofing 1.2 mixed waterproofing materials with concrete/plaster 1.3 performed waterproofing works 1.4 cleaned/maintained the workplace
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

Unit of Competency: UNDERSTAND MODERN MASONRY TECHNIQUES	Nominal Duration: 15 hrs.	Unit Code: SICIP-CON-AM-7-O
Unit descriptor: This unit covers the knowledge, skills and attitudes required to understand modern masonry techniques. It specifically includes the tasks of understanding ready mix casting techniques, describing auto plastering techniques and explaining auto brick laying techniques with robot.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand ready mix casting techniques	1.1 <u>Ready mix concreting process</u> is understood as a pre-mixed blend of cement, aggregates, and water. 1.2 Benefits of using ready-mix concrete are recognized. 1.3 Time constraints for using ready mix concrete are recognized. 1.4 <u>Pouring and casting techniques</u> for ready mix concrete are understood. 1.5 Importance of curing is understood. 1.6 Health and safety precautions related to working with ready mix concrete are understood.
2. Describe auto plastering techniques	2.1 Operation of auto plastering machines is understood. 2.2 Material selection for auto plastering is understood. 2.3 <u>Surface preparation techniques</u> are recognized. 2.4 The benefits of auto plastering are understood. 2.5 Cleaning and maintenance of auto plastering machines are understood.
3. Explain auto brick laying techniques with robot	3.1 <u>Auto brick laying robots</u> are understood. 3.2 The benefits of brick laying robots are understood. 3.3 Brick laying process is explained. 3.4 Mortar dispensing mechanisms are explained.

Range of Variables

Variable	Range
	May include but not limited to:
1. Ready mix concreting process	1.1 Sourcing and Receiving Raw Materials 1.2 Material Handling and Storage 1.3 Batching 1.4 Water Addition 1.5 Mixing 1.6 Transportation

	1.7 Placement and Compaction
2. Pouring and casting techniques	2.1 Tailgating 2.2 Wheelbarrows 2.3 Concrete Buggies 2.4 Pumping 2.5 Conveyor Trucks 2.6 Chuting and Wheeling 2.7 Formwork Removal and Weather Protection
3 Surface preparation techniques	3.1 Cleaning the Surface 3.2 Repairing and Filling 3.3 Moisture Control 3.4 Applying Bonding Agents (Optional) 3.5 Additional Considerations
4 Auto brick laying robots	4.1 Hadrian X 4.2 SAM100 (Semi-Automated Mason) 4.3 Tybot 4.4 MULE (Material Unit Lift Enhancer) 4.5 RoboBrix 4.6 BrickBot

Curricular Content Guide

1. Underpinning Knowledge	1.1 Ready-mix concreting process 1.2 Benefits of using ready-mix concrete in construction projects. 1.3 Time constraints and working time limitations for ready-mix concrete use. 1.4 Proper pouring and casting techniques. 1.5 Operation principles of auto plastering machines. 1.6 Suitable materials for use in auto plastering applications. 1.7 Function and working of auto brick laying robots. 1.8 Benefits and productivity improvements offered by brick laying robots. 1.9 Brick laying process including positioning and layering.
2. Underpinning Skills	2.1 Mixing ready-mix concrete 2.2 Pouring ready-mix concrete 2.3 Casting ready-mix concrete 2.4 Describing auto plastering machines 2.5 Preparing process surfaces for auto plastering 2.6 Maintaining procedure auto plastering machines 2.7 Explaining auto brick laying robots 2.8 Dispensing mechanism of mortar
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Tools and equipment

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood ready mix casting techniques. 1.2 described auto plastering techniques. 1.3 explained auto brick laying techniques with robot.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written Test 2.2 Practical Demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done at a training center. 3.2 Assessment should be done by the certified assessor of TÜV SÜD.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Advanced Masonry
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/Lab/Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S. N.	Major Equipment and Training facilities	Required facilities
1.	Computers/Laptop	01
2.	Projector & Screen (HD quality, HDMI support, 3000+ lumens)	01
3.	Printer	01
4.	Masonry Tools (Hammers, Trowels, Masonry Saws)	20
5.	Laser light/laser level	05
6.	Shovels	10
7.	Wooden float	20
8.	Aluminum floating roll	20
9.	Plumb-bob	20
10.	Sprit level	20
11.	Masonry tray	20
12.	Water level pipe	05
13.	Concrete mixing machine (electric and manual)	02
14.	Measuring Tape	20
15.	Pointing Trowels	10
16.	Trolly	02

Advanced Masonry

17.	Masonry Drill machine	02
18.	Mortar Mixing Buckets	20
19.	Try square	20
20.	Scaffolding with accessories	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on **Advanced Masonry**, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.